

Requirements-based testing of deep neural networks.

RBT4DNN · A critical review of its empirical evidence and two focused additional analyses.

SEMINAR

AI Testing Landscape: Ensuring Robustness through Advanced Testing Methods

COURSE LEADERSHIP

Prof. Dr. Alexander Pretschner
Supervisors: Vivek V. Vekariya · Simon Speth

PRESENTATION

Caspar Breloh
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Three questions guide the argument.

01 • NEED

Why do neural-network tests need requirements?

PROBLEM + BACKGROUND

02 • METHOD

How does RBT4DNN generate and evaluate targeted tests?

APPROACH + METHODOLOGY + RESULTS

03 • EVIDENCE

How convincing—and scalable—is the resulting evidence?

CRITIQUE + ADDITIONAL ANALYSES

High average accuracy does not answer a specific requirement.

CONVENTIONAL TESTING

Known behavior

A specification maps an input condition to an expected output. Test oracles can check the result directly.

DNN TESTING

Learned behavior

The model is inferred from examples. Global accuracy summarizes a distribution, not every meaningful scenario.

THE MISSING LINK

Semantic scenarios

Developers need evidence for statements such as “a very thick 7 must still be classified as 7.”

GLOBAL METRIC

How often is the complete test set correct?



SEMANTIC SLICE

Which rare inputs does the score contain?



REQUIREMENT VERDICT

Does the model satisfy this specific contract?

Existing generators rarely start from the requirement itself.

Neuron coverage, image transformations, and search-based generators can produce many tests—without proving that those tests represent the scenario a stakeholder cares about.

INPUT	Requirement	Structured natural language defines a semantic precondition and an expected outcome.
METHOD	RBT4DNN	Fine-tune a generator on images that satisfy that precondition.
OUTPUT	Targeted tests	Run the learned component and evaluate the requirement-specific oracle.

Four domains test whether one workflow survives increasingly ambiguous semantics.

MNIST · 7 REQUIREMENTS MNIST artifact samples Source-linked in interactive deck MNIST Measured height, thickness, and lean. A controlled starting point. Output: digit class	CELEBA-HQ · 7 REQUIREMENTS CelebA-HQ artifact samples Source-linked in interactive deck CelebA-HQ Ambiguous attributes labeled by visual question answering or humans. Output: eyeglasses decision	SGSM · 7 REQUIREMENTS SGSM artifact samples Source-linked in interactive deck SGSM Synthetic driving scenes with spatial relations linked to steering and acceleration. Output: numeric control	IMAGENET · 4 REQUIREMENTS ImageNet artifact samples Source-linked in interactive deck ImageNet Animal morphology linked to allowed classes in the WordNet taxonomy. Output: class set
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[1] Table 2 and §4.1.1-§4.1.3 · samples linked from the pinned public artifact

MNIST M3 artifact samples
Source-linked in interactive deck

A requirement defines both relevance and correctness.

If the digit is a 7 and is very thick, then the model shall label it as 7.

PRECONDITION

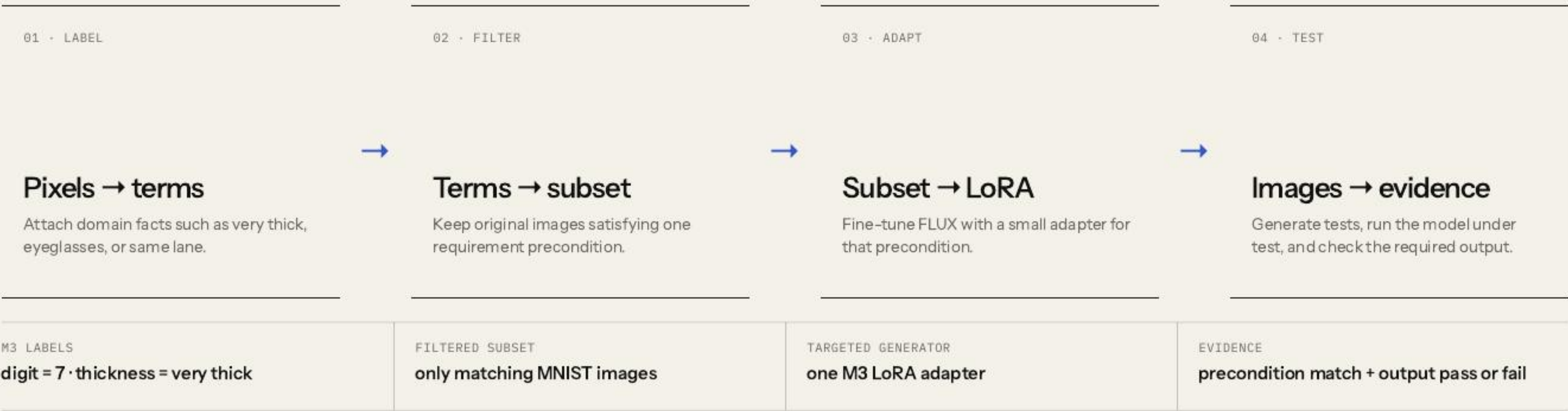
Which images count as relevant tests?

POSTCONDITION

Which model outputs count as acceptable?

[1] Table 2, M3 · a finite passing suite cannot prove the universal requirement

RBT4DNN turns one requirement into a targeted test campaign.



[1] Fig. 2 and §3 · symbols are translated into an explicit input/output contract

Glossary terms make visual requirements operational.

GLOSSARY TERM

A named visual property that can be assigned to an image: “very thick,” “same lane,” or “eyeglasses.”

01 · DEFINE

Authors specify the glossary

Requirement concepts are manually turned into a finite vocabulary of visual properties.

TERM CREATION IS NOT AUTOMATIC

02 · LABEL ORIGINAL DATA

Labels come from domain-specific signals

MNIST measurements · SGSM scene graphs · MiniCPM visual Q&A for CelebA-HQ and ImageNet.

CELEBA ALSO HAS HUMAN LABELS

03 · USE THE LABELS

Filter first; check later

Labels select the LoRA training subset. Separate binary classifiers then check glossary terms on generated images.

PRECONDITION MATCH · RQ1 / RQ3

Boundary: separate training labels and evaluation classifiers reduce direct reuse, but neither constitutes independent semantic ground truth.

[1] §3.2 and §4.1.6 · mean glossary-classifier accuracy 94.5%; C4 is 70.8%

One filtered precondition subset trains one targeted LoRA.

FROZEN BACKBONE

TRAINABLE UPDATE

W + BA

FLUX.1-dev: a pretrained 12-billion-parameter image generator.

Small low-rank matrices adapt the model to one filtered subset.

$$W' = W + BA$$

The original weights W stay frozen; inference uses the adapted weights W' after adding the learned low-rank update BA .

Training



Prompt



LoRA



Same S1 precondition • training reference (top) • prompt-only FLUX (middle) • targeted LoRA (bottom) • original paper Table 1 • CC BY 4.0.

The six research questions validate inputs before interpreting model outcomes.

A **credible failure is a joint event**: the generated input satisfies the precondition, but the model output violates the postcondition. RQ = Research Question.

Validate the generated inputs

RQ1–RQ3

- RQ1

Consistency

Does the image satisfy the precondition?
- RQ2

Plausibility

Is its distribution close to matching training images?
- RQ3

Variation

Are other glossary features distributed similarly?

Evaluate the resulting tests

RQ4–RQ6

- RQ4

Comparison

Are targeted outcomes more informative than baselines?
- RQ5

Violations

Do tests reveal credible postcondition failures?
- RQ6

Diagnosis

Do failures expose repeatable decision patterns?

[1] §4.1 · input validity is established before model outcomes are interpreted

Targeted LoRAs satisfy the requested precondition 92.1% of the time on average.

92.1%

mean across 21 requirements in MNIST, CelebA-HQ, and SGSM.

M3 · targeted LoRA

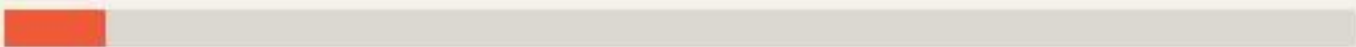
filtered images: very thick 7



95.8%

M3 · one all-data LoRA

all annotated MNIST training images; not plain FLUX



7.4%

COMPARISON SHOWN

Precondition-specific LoRA versus one LoRA trained on all annotated MNIST data · 100 images × 10 repetitions.

BOUNDARY

Glossary classifiers average 94.5% accuracy; they are measurement proxies, not semantic ground truth.

[1] §4.1.4, §4.2.1 and Fig. 3 · DeepHyperion and image transformations return on RQ4

KID compares each generator with real images satisfying the same precondition.

MNIST · M3 · KID ↓ .308 → .039 PROMPT-ONLY FLUX TARGETED LORA .308 .039	SGSM · S2 · KID ↓ .242 → .038 PROMPT-ONLY FLUX TARGETED LORA .242 .038	CELEBA-HQ · C1 · KID ↓ .123 → .097 PROMPT-ONLY FLUX TARGETED LORA .123 .097
KID · KERNEL INCEPTION DISTANCE A distance between two image sets in a learned feature space · lower means the distributions are closer.	BOUNDARY KID is not an image-level realism probability; several reference subsets contain fewer than 300 images.	

[1] §4.2.2 and Table 3 · examples show KID(reference, prompt-only FLUX) → KID(reference, targeted LoRA)

Targeted LoRA matches labeled feature frequencies— but RQ3 cannot show how much visual variation remains.

MNIST · MEAN JS 7.2× closer PROMPT-ONLY FLUX .1765 TARGETED LORA .0244	CELEBA-HQ · MEAN JS 1.4× closer PROMPT-ONLY FLUX .1150 TARGETED LORA .0831	SGSM · MEAN JS 5.5× closer PROMPT-ONLY FLUX .1960 TARGETED LORA .0355
WHAT “FEATURE FREQUENCY” MEANS Example: among thick 7s, how often are images also left-leaning, right-leaning, tall, or short? Compare those label proportions with the filtered training subset.	JS · JENSEN-SHANNON DIVERGENCE One distance between those two label distributions · lower means closer. It does not measure pixel diversity, memorization, or latent-space coverage.	

[1] §4.2.3 and Table 4 · displayed domain means are calculated from the seven reported rows

M2 requirement: if the image is a **very thick 3**, the classifier must predict 3.

Technique	Passed Tests	Failed Tests
Deephyperion		
Image Transform		
RBT4DNN		

Original paper evidence · Table 5 · cropped only for legibility · CC BY 4.0.

WHAT PASS / FAIL MEANS HERE

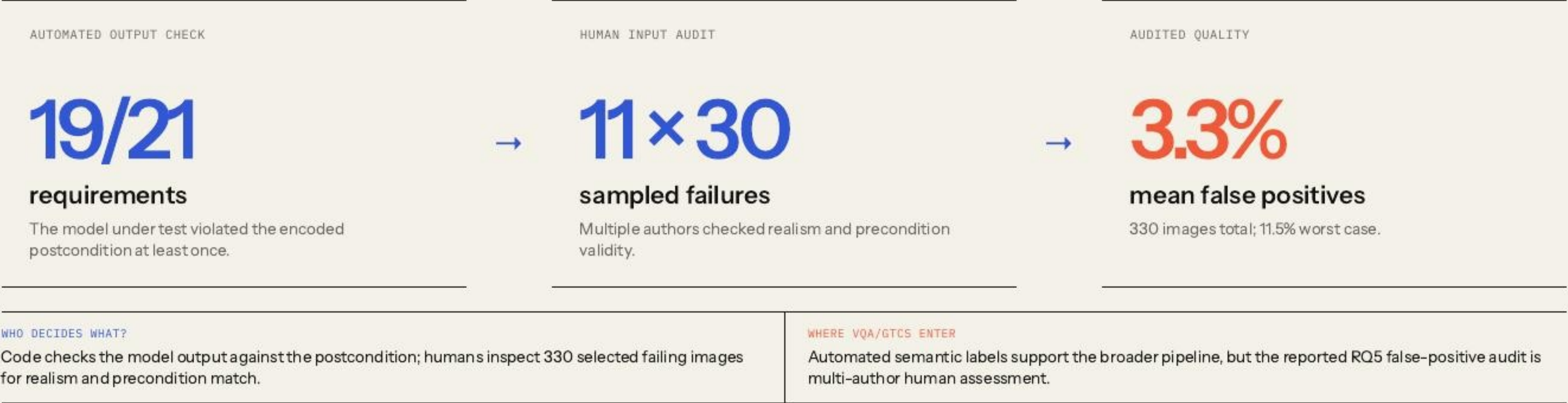
For M2, pass = the MNIST classifier predicts “3”; fail = it predicts another digit. The rows separate those outputs for each generator.

INTERPRETATION RULE

DeepHyperion and transformations mostly miss “very thick”; therefore neither their passes nor failures are evidence about M2.

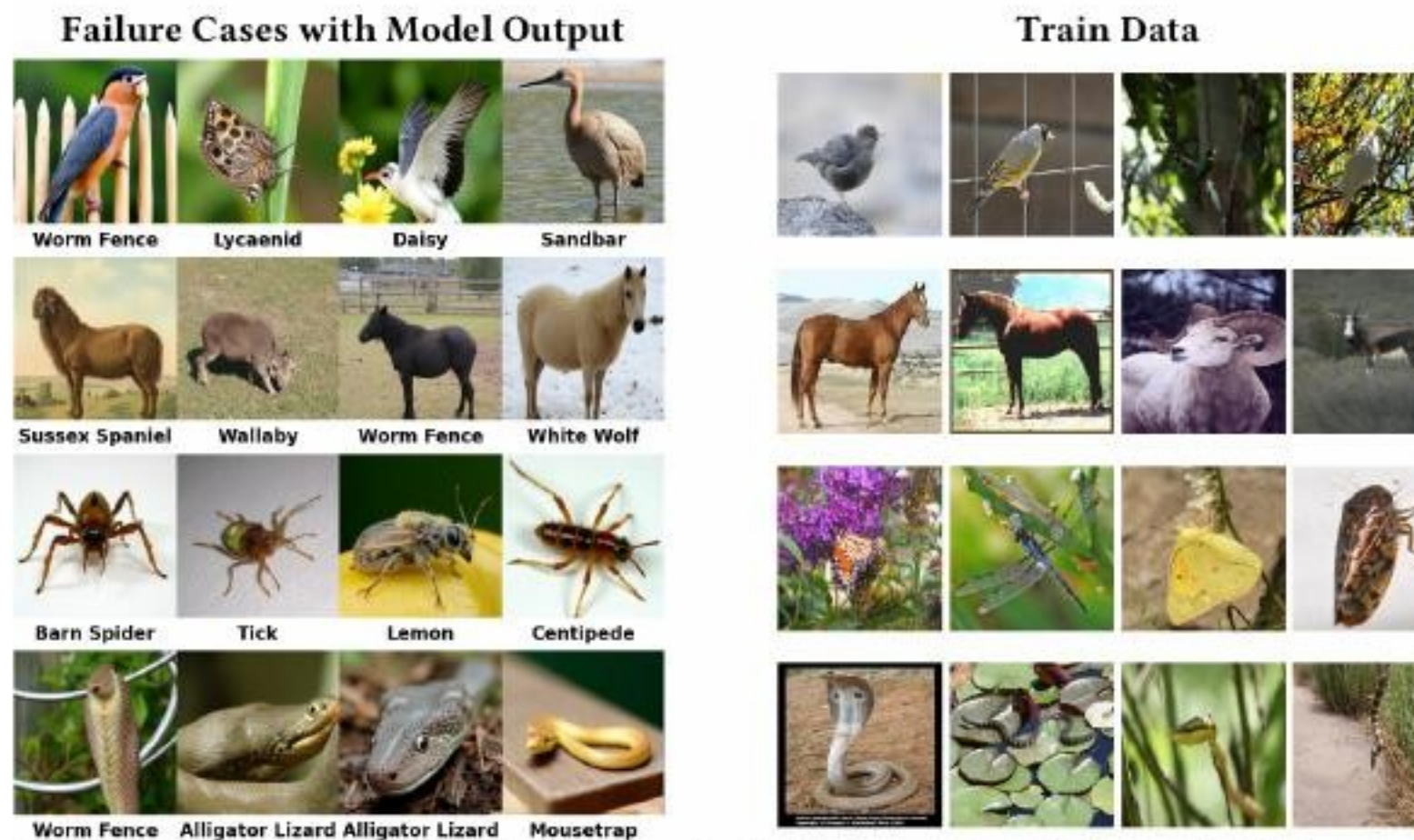
[1] §4.3.1 and Table 5 · reproduced from Mozumder et al. under CC BY 4.0

RBT4DNN exposed failures broadly, and the audited subset was mostly valid.



[1] §4.1.6 and §4.3.2 · assessor disagreement on precondition consistency was 1.7%

Across four animal requirements and three classifiers, failures reveal recurring model-specific patterns.



Original paper Table 7 · generated failure cases and training examples · cropped for legibility · CC BY 4.0.

METHOD

4 requirements × 3 classifiers × repeated generation

Collect 1,486 postcondition failures, rank frequent wrong labels, then manually inspect associated images.

EXAMPLE

A bird near a fence → “worm fence”

Other cases align with insects, horses, or snakes that resemble frequent training classes or backgrounds.

BOUNDARY

Association, not causal diagnosis

The study does not change only the background while holding the animal fixed.

The paper demonstrates feasibility—not a complete assurance process.

GENERATIVE TRAINING Better tests Improve generator training and adaptation to reveal more observed failures and reduce false positives.	COVERAGE EVIDENCE Latent-space coverage Map the model's internal feature space and deliberately search regions that generated tests have not exercised.	GENERALIZATION More domains Expand driving-like safety datasets and explore broader robustness and functional requirements.
ASSURANCE BOUNDARY The comparator changes with the question; automated labels, narrow baselines, and small KID subsets limit how far the evidence generalizes.		

[1] §4.4 and §5 · author-stated scope kept separate from this seminar's critique

A credible failure needs two things to be true.

INPUT		OUTPUT		EVIDENCE
Precondition is true The generated image really represents the requested scenario.		+	Postcondition is false The tested neural network violates the expected behavior.	Credible counterexample A relevant input exposes a requirement-specific model error.
MOST RESULTS			STRONGEST EVIDENCE	
Input validity is automated Glossary classifiers are useful—but imperfect—semantic proxies.			330 failures are manually audited Confidence is strongest for this selected sample, not automatically for every reported failure.	

Sharing one MNIST LoRA saves training—but sharply weakens targeting on the hardest requirements.

METHOD	EASY TO SHARE · M4	HARD TO SHARE · M3
Reuse the authors' MNIST ablations Compare precondition-match rates for a targeted LoRA, one LoRA over six requirement subsets, and one LoRA over all MNIST data. 6 requirements	Targeted 92.5% · shared 56.4% The shared requirement LoRA remains usable, though substantially less precise. The all-data LoRA reaches only 12.7%. training ↘ · targeting ↘	Targeted 95.8% · shared 27.5% The all-data LoRA reaches 7.4%. A single generator does not preserve difficult semantic regions automatically. targeting matters

EXTENSION MESSAGE

Per-requirement LoRAs are not merely an implementation detail: they buy semantic precision. The open design question is whether conditioning can recover that precision in one shared model.

Sharing can cut assumed training compute by 75%—but loses the targeting that makes tests relevant.

6 TARGETED LORAS One adapter per requirement Six assumed 1-hour training runs. Each adapter sees only images matching its own precondition. \$18 · M3 95.8%	1 SHARED-REQUIREMENTS LORA Union of six filtered subsets One assumed 1.5-hour run. Cheaper, but the requirement signal competes inside one adapter. \$4.50 · M3 27.5%	1 ALL-DATA LORA All annotated MNIST data One assumed 1.5-hour run. Cheapest model count, but no requirement-specific filtering. \$4.50 · M3 7.4%
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DEFENSIBLE USE OF THE MODEL

Targeting rates come from the authors’ ablations; costs are scenario assumptions, not invoices. Absolute “cost per valid failure” is not claimed—the model is useful only for relative prioritization under identical assumptions.

[1] Fig. 3 · seminar mnist-lora-ablation and cost-analysis · illustrative training assumptions documented in the artifact

Requirements make tests **relevant**. Validity makes failures **credible**.

WHAT THE PAPER ESTABLISHES

Structured requirements can guide distributionally plausible, glossary-diverse, scenario-specific test generation across four domains.

WHAT LIMITS THE CLAIM

Semantic labels and glossary classifiers are useful proxies; credible failures still depend on joint input-and-output validation.

References.

[1]

N. J. Mozumder, F. Toledo, S. Dola, and M. B. Dwyer. “RBT4DNN: Requirements-based Testing of Neural Networks.” arXiv:2504.02737, v3, 2025. CC BY 4.0.

[2]

E. J. Hu et al. “LoRA: Low-Rank Adaptation of Large Language Models.” ICLR, 2022.

[3]

RBT4DNN artifact. github.com/less-lab-uva/RBT4DNN · seminar provenance pinned to commit c3deff871ed41043d83a05289faab84c41706586.

[4]

Seminar experiments. Reproducible notebooks, results, summaries, and provenance in this repository.

Original paper ↗	Original artifact ↗
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